

CardFlow Credit Origination Platform

Multi-Tenant SaaS Solution for Credit Card
Application and Approval Automation

System Requirement Specification Document

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Version History

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1. Introduction

This **Software Requirements Specification (SRS)** defines the system-level requirements for the **CardFlow Credit Origination Platform**, a multi-tenant SaaS solution designed to support the end-to-end lifecycle of credit card application processing.

The document describes the functional and non-functional requirements of the platform, including system behavior, data handling, workflow processing, and external interactions required to support credit application submission, review, approval, amendment, and reinstatement processes.

This SRS serves as the primary reference for system design, development, testing, and validation activities.

1.1. Purpose of the Document

The purpose of this document is to provide a detailed specification of system requirements that define how the **CardFlow Credit Origination Platform** operates.

This document aims to:

- Define system behavior and functional capabilities
- Establish detailed requirements for system interactions and workflows
- Provide a foundation for technical design and implementation
- Support traceability between business requirements, system design, and testing

1.2. Product Overview

The **CardFlow Credit Origination Platform** enables financial institutions to digitize and standardize the credit card application lifecycle through a configurable, role-based workflow system.

The platform supports:

- Credit application submission by applicants or authorized tenant users
- Multi-level review and approval workflows
- Configurable approval routing based on defined business rules
- Document management and audit logging
- Credit amendment and reinstatement processes
- Administrative configuration of roles, permissions, and approval matrices

The platform operates in a multi-tenant SaaS environment, allowing each tenant organization to configure workflows and operational rules independently while maintaining data isolation.

1.3. Scope of the System

The system supports the management of credit card application processes within tenant organizations, including submission, validation, review, approval, amendment, and reinstatement activities.

The scope of the system includes:

- Management of applicant data and supporting documentation
- Execution of configurable approval workflows
- Enforcement of business rules related to submission and eligibility
- Tracking of application status throughout the lifecycle
- Recording of audit logs for all system and user activities

The system does not include external credit scoring computation, core banking transaction processing, or card issuance functions.

1.4. Intended Audience

This document is intended for stakeholders involved in the design, development, testing, and implementation of the **CardFlow Credit Origination Platform**, including:

- Business Analysts
- System Architects
- Software Developers
- Quality Assurance Engineers
- Technical Writers
- Project Stakeholders

1.5. Definitions and Terminology

Key terms used in this document are defined in the **Business Requirements Document** (BRD) and are referenced to maintain consistency across documentation artifacts.

1.6. References

- CardFlow Credit Origination Platform – Business Requirements Document
- Internal System Design and Architecture References (TBD)

Sample Text

2. System Overview

This section provides an overview of the **CardFlow Credit Origination Platform** from a system perspective. It describes the system's operating context, key user roles, supported workflows, and the environment in which the platform operates. This overview establishes a foundational understanding of how the system interacts with users and external components before detailing specific system requirements.

2.1. System Context

The **CardFlow Credit Origination Platform** is a multi-tenant SaaS application that supports the end-to-end lifecycle of credit card application processing within tenant organizations. The platform:

- Enables interaction between multiple user roles, including applicants, tenant users (such as sales officers, reviewers, and approvers), and administrative users. These users interact with the system through a centralized interface to submit applications, review information, make approval decisions, and manage system configurations.
- Operates as a centralized system that manages application data, workflow processing, document storage, and audit tracking, while allowing tenant organizations to configure their own approval hierarchies and operational rules.

2.2. Business Process Summary

The system supports a structured workflow that governs the lifecycle of credit applications from initiation to outcome.

Core workflows supported by the system include:

- Credit Application Processing
- Credit Amendment Processing
- Credit Reinstatement Processing

These workflows follow a consistent lifecycle pattern:

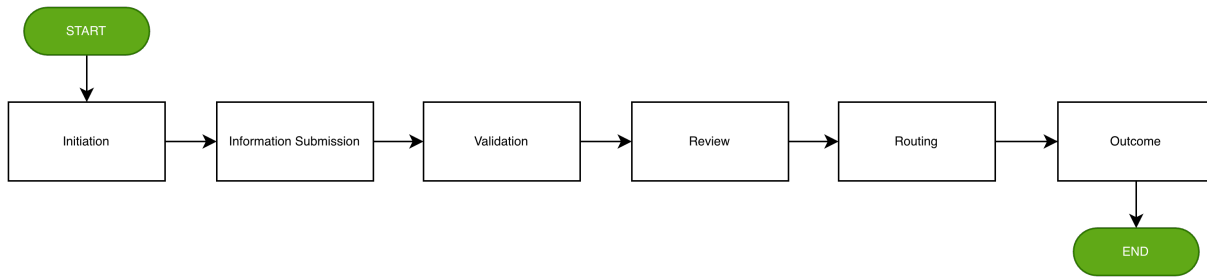


Figure 1 CardFlow Credit Origination Platform Workflow Lifecycle

Each stage represents a logical step in the processing of credit-related requests and is governed by system rules, validation logic, and configurable workflow parameters.

2.3. User Classes and Characteristics

The system supports multiple user classes, each with defined roles and responsibilities within the credit origination process.

Table 1 User Classes and Characteristics

Term	Definition
Applicant	External users who submit credit card applications and provide required supporting documentation.
Sales Officer	Authorized tenant users who assist applicants and may initiate or manage application submissions.
Risk Reviewer	Tenant users responsible for evaluating application data and supporting documents to assess credit risk.
Compliance Reviewer	Tenant users responsible for verifying regulatory and documentation requirements prior to approval.
Approver	Authorized personnel responsible for making final approval or rejection decisions based on defined approval thresholds.

Term	Definition
Platform Administrator	Tenant users responsible for configuring system settings, managing roles and permissions, and maintaining approval matrices.

2.4. Operating Environment

The **CardFlow Credit Origination Platform** operates as a cloud-based, multi-tenant SaaS solution accessible through web-based interfaces.

The system supports:

- Browser-based access for all user roles
- Secure authentication mechanisms for both external and internal users
- Centralized data storage and processing
- Configurable workflows and role-based access control

The platform is designed to support multiple tenant organizations operating independently within a shared system environment.

2.5. Assumptions and Dependencies

The operation of the system is based on several assumptions and external dependencies.

Key assumptions include:

- Tenant organizations define and configure their approval workflows and business rules.
- Users are trained to operate the platform according to their assigned roles.
- Required infrastructure and network connectivity are available for system access.

Key dependencies include:

- Availability of authentication and user management services
- Integration with external systems providing customer or credit-related data (if applicable)
- Timely configuration of approval matrices and operational parameters by tenant administrators

Sample

3. Functional Requirements

This section defines the functional requirements of the **CardFlow Credit Origination Platform**. It describes the system capabilities and behaviors required to support the end-to-end credit application lifecycle, including user interactions, workflow processing, document handling, and administrative configuration.

The requirements are organized into functional areas aligned with the core processes supported by the system. Each requirement specifies a system behavior that must be implemented to satisfy the operational needs defined in the **Business Requirements Document**.

Note:

Each functional requirement is uniquely identified and traceable to corresponding business requirements through **Requirements Traceability Matrix** (RTM).

3.1. Authentication and Access Control

This subsection defines the requirements governing user authentication, session management, and access control within the **CardFlow Credit Origination Platform**. These requirements ensure only authorized users can access the system and perform actions according to their assigned roles and permissions.

Table 2 Authentication and Access Control

No.	Functional Requirement
FR-AUTH-01	The system shall allow applicants to authenticate using registered credentials to access and manage credit applications.
FR-AUTH-02	The system shall allow tenant users to authenticate securely before accessing system functionalities related to application processing, review, approval, and administration.
FR-AUTH-03	The system shall enforce role-based access control to ensure that users can only perform actions permitted by their assigned roles.
FR-AUTH-04	The system shall restrict access to sensitive application data and administrative functions to authorized users only.

No.	Functional Requirement
FR-AUTH-05	The system shall maintain authenticated user sessions for the duration of user activity.
FR-AUTH-06	The system shall automatically terminate inactive user sessions after a defined period of inactivity.
FR-AUTH-07	The system shall allow authorized administrative users to create, update, and deactivate user accounts.
FR-AUTH-08	The system shall allow administrative users to assign and manage user roles and permissions within the tenant environment.
FR-AUTH-09	The system shall prevent unauthorized access attempts and enforce authentication validation before granting system access.

3.2. Credit Application Submission

This subsection defines the requirements governing the creation, validation, and submission of credit card applications within the platform. These requirements ensure that applications are submitted with complete and valid information and are properly prepared for downstream review and approval processes.

Table 3 Credit Application Submission

No.	Functional Requirement
FR-APP-01	The system shall allow authorized users to create a new credit card application.
FR-APP-02	The system shall enforce a rule that prevents the creation of multiple active applications for the same applicant.
FR-APP-03	The system shall require applicants to provide mandatory information prior to submission.
FR-APP-04	The system shall allow applicants to upload supporting documents required for credit evaluation.

No.	Functional Requirement
FR-APP-05	The system shall support predefined file formats for uploaded documents.
FR-APP-06	The system shall enforce file size limits for uploaded documents.
FR-APP-07	The system shall validate that all required information and supporting documents are provided before allowing submission.
FR-APP-08	The system shall prevent submission of applications that fail validation checks.
FR-APP-09	The system shall allow authorized users to submit completed applications for review.
FR-APP-10	The system shall prevent modification of submitted applications unless returned for correction.
FR-APP-11	The system shall assign a unique identifier to each submitted application.
FR-APP-12	The system shall maintain and update application status throughout the lifecycle of the application.

3.3. Document Management

This subsection defines the requirements related to the handling, storage, and accessibility of documents associated with credit applications. These requirements ensure that documents are securely stored, properly linked to application records, and available for review by authorized users.

Table 4 Document Management

No.	Functional Requirement
FR-DOC-01	The system shall associate uploaded documents with the corresponding credit application record.

No.	Functional Requirement
FR-DOC-02	The system shall allow authorized users to access and review documents associated with an application.
FR-DOC-03	The system shall allow documents to be categorized based on predefined document types.
FR-DOC-04	The system shall preserve the integrity of uploaded documents without altering their original content.
FR-DOC-05	The system shall allow reviewers to validate submitted documents as part of the credit evaluation process.
FR-DOC-06	The system shall retain documents for audit and reference purposes.

3.4. Risk Review and Approval Workflow

This subsection defines the requirements governing the review and approval workflow of credit applications within the **CardFlow Credit Origination Platform**. It describes how applications are evaluated, routed, and processed through configurable approval hierarchies to ensure that credit decisions are made in accordance with defined business rules and organizational policies.

The workflow supports multiple levels of review and approval, enabling tenant organizations to implement structured decision-making processes for credit evaluation.

Table 5 Risk Review and Approval Workflow

No.	Functional Requirement
FR-REV-01	The system shall allow authorized reviewers to access submitted credit applications for evaluation.
FR-REV-02	The system shall allow reviewers to view application details and associated supporting documents.
FR-REV-03	The system shall allow reviewers to record evaluation outcomes and remarks during the review process.

No.	Functional Requirement
FR-REV-04	The system shall allow reviewers to request additional information by returning the application to the applicant or initiating user.
FR-REV-05	The system shall route applications to the appropriate reviewer or approval stage based on configured workflow rules.
FR-REV-06	The system shall support multi-level review and approval workflows based on tenant-defined approval hierarchies.
FR-REV-07	The system shall determine approvers based on predefined criteria such as approval thresholds, roles, or organizational hierarchy.
FR-REV-08	The system shall allow approvers to approve, reject, or return applications.
FR-REV-09	The system shall require approvers to provide remarks when rejecting or returning an application.
FR-REV-10	The system shall update the application status based on review and approval decisions.
FR-REV-11	The system shall maintain a record of all review and approval actions performed on an application.
FR-REV-12	The system shall prevent further processing of applications that are rejected.
FR-REV-13	The system shall allow returned applications to be modified and resubmitted by authorized users.

3.5. Credit Amendment Process

This subsection defines the requirements governing the amendment of existing credit applications or credit account information within the **CardFlow Credit Origination Platform**. It describes how amendment requests are initiated, validated, and processed through configurable approval workflows.

The amendment process ensures that changes to previously submitted or approved credit information are controlled, properly reviewed, and authorized in accordance with defined business rules and approval hierarchies.

Table 6 Credit Amendment Process

No.	Functional Requirement
FR-AMD-01	The system shall allow authorized users to initiate amendment requests for existing credit applications or credit account records.
FR-AMD-02	The system shall validate whether a credit application or account is eligible for amendment based on defined business rules.
FR-AMD-03	The system shall prevent amendment requests for applications or accounts that are not eligible for modification.
FR-AMD-04	The system shall allow users to provide updated information and upload supporting documents as part of the amendment request.
FR-AMD-05	The system shall validate that all required amendment information and supporting documents are provided prior to submission.
FR-AMD-06	The system shall prevent submission of amendment requests that fail validation checks.
FR-AMD-07	The system shall route amendment requests through the configured approval workflow to authorized reviewers and approvers.
FR-AMD-08	The system shall allow reviewers and approvers to access amendment request details and associated supporting documents.
FR-AMD-09	The system shall allow approvers to approve, reject, or return amendment requests.

No.	Functional Requirement
FR-AMD-10	The system shall require approvers to provide remarks when rejecting or returning an amendment request.
FR-AMD-11	The system shall update the status of the amendment request based on the approval decision.
FR-AMD-12	The system shall maintain a record of all actions performed on amendment requests for audit purposes.
FR-AMD-13	The system shall allow returned amendment requests to be modified and resubmitted by authorized users.

3.6. Credit Reinstatement Process

This subsection defines the requirements governing the reinstatement of previously suspended or inactive credit accounts within the **CardFlow Credit Origination Platform**. It describes how reinstatement requests are initiated, validated, and processed through the configured approval workflow.

The reinstatement process ensures that reactivation of credit accounts is controlled, evaluated, and authorized in accordance with defined eligibility criteria and organizational policies.

Table 7 Credit Reinstatement Process

No.	Functional Requirement
FR-RST-01	The system shall allow authorized tenant users to initiate reinstatement requests for suspended or inactive credit accounts.
FR-RST-02	The system shall validate whether a credit account is eligible for reinstatement based on defined eligibility criteria.
FR-RST-03	The system shall prevent reinstatement requests for accounts that do not meet eligibility conditions.
FR-RST-04	The system shall allow users to provide supporting documents and updated information required for reinstatement.

No.	Functional Requirement
FR-RST-05	The system shall validate that all required information and supporting documents are provided prior to submission.
FR-RST-06	The system shall prevent submission of reinstatement requests that fail validation checks.
FR-RST-07	The system shall route reinstatement requests through the configured approval workflow to authorized reviewers or approvers.
FR-RST-08	The system shall allow reviewers or approvers to access reinstatement request details and associated supporting documents.
FR-RST-09	The system shall allow approvers to approve or reject reinstatement requests.
FR-RST-10	The system shall require approvers to provide remarks when rejecting a reinstatement request.
FR-RST-11	The system shall update the status of the reinstatement request based on the approval decision.
FR-RST-12	The system shall maintain a record of all actions performed on reinstatement requests for audit purposes.

3.7. Reporting and Audit Capabilities

This subsection defines the requirements related to reporting and audit capabilities within the **CardFlow Credit Origination Platform**. It describes how the system provides visibility into application data, workflow activities, and user actions, as well as how it maintains audit records for traceability and compliance purposes.

These capabilities enable tenant organizations to monitor application progress, generate operational insights, and ensure accountability through comprehensive audit tracking.

Table 8 Reporting and Audit Capabilities

No.	Functional Requirement
FR-RPT-01	The system shall allow authorized users to view application status and progress throughout the application lifecycle.
FR-RPT-02	The system shall allow users to retrieve application details and associated workflow history.
FR-RPT-03	The system shall allow authorized users to generate reports based on application data, status, and processing outcomes.
FR-RPT-04	The system shall allow reports to be filtered based on predefined criteria such as date range, application status, and user role.
FR-RPT-05	The system shall provide visibility into review and approval activities associated with each application.
FR-RPT-06	The system shall maintain audit logs that record all user and system actions performed on applications.
FR-RPT-07	The system shall capture audit details including user identity, action performed, timestamp, and affected records.
FR-RPT-08	The system shall allow authorized users to access audit logs for review and compliance purposes.
FR-RPT-09	The system shall ensure that audit logs are preserved and cannot be altered by unauthorized users.
FR-RPT-10	The system shall support retrieval of audit data for reporting and investigation purposes

3.8. Administrative Configuration

This subsection defines the requirements related to administrative configuration within the **CardFlow Credit Origination Platform**. It describes how authorized administrative users manage system settings, user roles, permissions, and workflow configurations to support tenant-specific operational requirements.

These capabilities enable tenant organizations to customize system behavior, enforce access control, and define approval structures in accordance with their internal policies and processes.

Table 9 Administrative Configuration

No.	Functional Requirement
FR-ADM-01	The system shall allow authorized administrative users to configure user roles and permissions within the tenant environment.
FR-ADM-02	The system shall allow administrative users to assign roles to users and define access levels for system functionalities.
FR-ADM-03	The system shall enforce role-based access control based on configured roles and permissions.
FR-ADM-04	The system shall allow administrative users to configure approval workflows for credit application processing.
FR-ADM-05	The system shall allow administrative users to define approval hierarchies based on roles, thresholds, or organizational structure.
FR-ADM-06	The system shall allow administrative users to modify workflow configurations without requiring system code changes.
FR-ADM-07	The system shall allow administrative users to configure validation rules and operational parameters related to application processing.
FR-ADM-08	The system shall allow administrative users to manage lookup values and reference data used within the system.
FR-ADM-09	The system shall allow administrative users to activate or deactivate system configurations as required.
FR-ADM-10	The system shall maintain audit records of all administrative configuration changes.

3.9. Notification and Communication Requirements

This subsection defines the requirements related to system-generated notifications and communication within the **CardFlow Credit Origination Platform**. It describes how the system informs users of important events, status changes, and required actions throughout the credit application lifecycle.

These requirements ensure that all relevant stakeholders are appropriately notified during key stages of processing, enabling timely actions and improving workflow efficiency.

Table 10 Notification and Communication Requirements

No.	Functional Requirement
FR-NOT-01	The system shall generate notifications for key events in the application lifecycle, including submission, review, approval, rejection, and return for additional information.
FR-NOT-02	The system shall notify applicants or initiating users when an application or request is returned for correction or additional information.
FR-NOT-03	The system shall notify reviewers and approvers when applications or requests are assigned to them for action.
FR-NOT-04	The system shall notify users of final outcomes, including approval or rejection of applications, amendment requests, and reinstatement requests.
FR-NOT-05	The system shall support configurable notification triggers based on workflow events.
FR-NOT-06	The system shall support delivery of notifications through predefined communication channels.
FR-NOT-07	The system shall include relevant application details in notifications to provide context for required user actions.
FR-NOT-08	The system shall ensure that notifications are sent in a timely manner following triggering events.

No.	Functional Requirement
FR-NOT-09	The system shall maintain a record of notifications sent for audit and reference purposes.

4. External Interface Requirements

This section describes the interfaces through which the **CardFlow Credit Origination Platform** interacts with users and external systems. It defines the user interface characteristics, system integration points, and communication mechanisms required to support application processing, workflow execution, and data exchange.

These interfaces ensure that users can effectively interact with the system and that the platform can integrate with external services where applicable.

4.1. User Interfaces

This subsection defines the characteristics and requirements of the user interface provided by the **CardFlow Credit Origination Platform**. It describes how users interact with the system through web-based screens and forms to perform tasks such as application submission, review, approval, and administration.

The user interface is designed to support role-based interactions, ensuring that users are presented with relevant functionalities based on their assigned roles and responsibilities.

Table 11 User Interfaces

No.	Functional Requirement
UI-01	The system shall provide a web-based user interface accessible through supported browsers.
UI-02	The system shall present role-based views that display functionalities and data relevant to the user's assigned role.
UI-03	The system shall provide input forms for capturing application data and supporting information.
UI-04	The system shall display validation messages for incomplete or invalid inputs.
UI-05	The system shall allow users to view application status and progress through dashboards or status indicators.
UI-06	The system shall provide interfaces for reviewing application details and associated documents.

No.	Functional Requirement
UI-07	The system shall provide interfaces for performing approval actions, including approve, reject, and return.
UI-08	The system shall support navigation between system modules such as application management, reporting, and administration.
UI-09	The system shall ensure that sensitive information is displayed only to authorized users based on access permissions.

4.2. System Interfaces

This subsection defines the interfaces between the **CardFlow Credit Origination Platform** and external or internal systems. These interfaces enable the exchange of data required to support credit application processing, validation, and operational workflows.

Table 12 System Interfaces

No.	Functional Requirement
SI-01	The system shall support integration with external systems for retrieval or validation of customer-related data, where applicable.
SI-02	The system shall support data exchange with external services through defined interface mechanisms.
SI-03	The system shall ensure that data exchanged with external systems is validated prior to processing.
SI-04	The system shall handle failures in external system communication and provide appropriate error handling mechanisms.
SI-05	The system shall ensure secure transmission of data between the platform and external systems.

4.3. API / Integration Interfaces

This subsection defines the application programming interfaces (APIs) and integration mechanisms supported by the **CardFlow Credit Origination Platform**. These interfaces enable external systems and services to interact with the platform for data exchange, workflow integration, and operational support.

Table 13 User Interfaces

No.	Functional Requirement
API-01	The system shall expose APIs to support creation, retrieval, and update of credit application data.
API-02	The system shall support secure API access through authentication and authorization mechanisms.
API-03	The system shall validate all incoming API requests before processing.
API-04	The system shall return appropriate response messages and status codes for API requests.
API-05	The system shall ensure data consistency between internal processing and external API interactions.

4.4. Communication Interfaces

This subsection defines the communication mechanisms used by the system to deliver notifications and exchange information with users and external services. These mechanisms support timely communication of workflow events and system updates.

Table 14 Communication Interfaces

No.	Functional Requirement
CI-01	The system shall support communication channels for delivering notifications to users.
CI-02	The system shall ensure that notifications are delivered securely and reliably.

No.	Functional Requirement
CI-03	The system shall support configurable communication settings based on tenant requirements.
CI-04	The system shall maintain records of communication events for audit and reference purposes.

5. Data Requirements

This section defines the data-related requirements of the **CardFlow Credit Origination Platform**. It describes the key data entities, data inputs, validation rules, storage considerations, and audit data management required to support the credit application lifecycle.

These requirements ensure that data is accurately captured, validated, stored, and maintained throughout the system while supporting traceability and compliance needs.

5.1. Core Business Entities

This subsection identifies the primary data entities managed by the system. These entities represent the core business objects required to support credit application processing, workflow management, and audit tracking.

The system manages the following key entities:

- **Applicant** – Represents individuals or organizations applying for credit
- **Credit Application** – Represents the submitted credit request and its associated data
- **Document** – Represents supporting files submitted as part of an application
- **User** – Represents system users including applicants, reviewers, approvers, and administrators
- **Workflow Instance** – Represents the progression of an application through defined workflow stages
- **Audit Record** – Represents logged system and user actions for traceability

5.2. Data Inputs

This subsection defines the types of data inputs required by the system to support credit application processing and related workflows. It describes how user-provided data and supporting information are captured and utilized within the platform.

Table 15 Data Input

No.	Functional Requirement
DATA-IN-01	The system shall capture applicant-provided information required for credit application processing.

No.	Functional Requirement
DATA-IN-02	The system shall capture supporting documents submitted by users as part of application, amendment, and reinstatement processes.
DATA-IN-03	The system shall capture user actions and inputs during review and approval activities.
DATA-IN-04	The system shall allow administrative users to input configuration data such as roles, permissions, and workflow parameters.

5.3. Data Validation Requirements

This subsection defines the validation requirements applied to data within the system. These validations ensure data completeness, accuracy, and consistency before processing and storage.

Table 16 Data Validation Requirements

No.	Functional Requirement
DATA-VAL-01	The system shall validate that all mandatory fields are completed prior to submission.
DATA-VAL-02	The system shall validate the format of input data based on predefined rules.
DATA-VAL-03	The system shall validate uploaded documents for supported formats and size constraints.
DATA-VAL-04	The system shall prevent processing of data that fails validation checks.
DATA-VAL-05	The system shall provide feedback to users for validation errors encountered during data entry or submission.

5.4. Data Storage and Retention

This subsection defines how data is stored, managed, and retained within the system. It ensures that data is securely maintained and available for operational, audit, and compliance purposes.

Table 17 Data Storage and Retention

No.	Functional Requirement
DATA-ST-01	The system shall store credit application data and associated records in a centralized data repository.
DATA-ST-02	The system shall store supporting documents in a secure and accessible storage system.
DATA-ST-03	The system shall ensure data integrity during storage and retrieval operations.
DATA-ST-04	The system shall retain data in accordance with defined retention policies.
DATA-ST-05	The system shall ensure that stored data is accessible to authorized users only.

5.5. Audit Data and Logs

This subsection defines the requirements for capturing and maintaining audit data within the system. Audit data provides traceability of user actions and system events, supporting compliance, monitoring, and investigation activities.

Table 18 Audit Data and Logs

No.	Functional Requirement
DATA-AUD-01	The system shall record all user actions and system events related to application processing.
DATA-AUD-02	The system shall capture audit details including user identity, action performed, timestamp, and affected data.
DATA-AUD-03	The system shall ensure that audit records are securely stored and protected from unauthorized modification.

No.	Functional Requirement
DATA-AUD-04	The system shall allow authorized users to access audit logs for monitoring and investigation purposes.
DATA-AUD-05	The system shall retain audit records in accordance with defined audit and compliance requirements.

6. Non-Functional Requirements

This section defines the non-functional requirements of the **CardFlow Credit Origination Platform**. These requirements describe the quality attributes and operational characteristics of the system, including performance, security, scalability, availability, usability, and compliance.

These requirements ensure that the system operates reliably, securely, and efficiently within a multi-tenant SaaS environment while meeting organizational and regulatory expectations.

6.1. Performance Requirements

This subsection defines the performance expectations of the system, including response times, processing efficiency, and system throughput during normal and peak usage conditions.

Table 19 Performance Requirements

No.	Functional Requirement
NFR-PERF-01	The system shall support concurrent access by multiple users without degradation of core functionalities.
NFR-PERF-02	The system shall process user actions such as application submission and status updates within acceptable response times.
NFR-PERF-03	The system shall support timely retrieval of application data and reports.
NFR-PERF-04	The system shall handle peak usage volumes without significant performance degradation.

6.2. Availability Requirements

This subsection defines the availability requirements of the system to ensure continuous access and minimal disruption to users.

Table 20 Availability Requirements

No.	Functional Requirement
NFR-AVL-01	The system shall be available to users during defined operational hours.

No.	Functional Requirement
NFR-AVL-02	The system shall minimize downtime during maintenance or system updates.
NFR-AVL-03	The system shall support recovery from system failures within an acceptable timeframe.

6.3. Security Requirements

This subsection defines the security requirements of the system to ensure the protection of sensitive data and prevention of unauthorized access.

Table 21 Security Requirements

No.	Functional Requirement
NFR-SEC-01	The system shall enforce secure authentication mechanisms for all users.
NFR-SEC-02	The system shall enforce role-based access control for all system functionalities.
NFR-SEC-03	The system shall ensure secure transmission of data between users and the system.
NFR-SEC-04	The system shall protect sensitive data from unauthorized access and disclosure.
NFR-SEC-05	The system shall log and monitor security-related events for detection and investigation purposes.

6.4. Privacy and Data Protection Requirements

This subsection defines the requirements related to the protection of personal and sensitive data handled by the system. These requirements ensure that data is processed and stored in accordance with applicable privacy standards and organizational policies.

Table 22 Privacy and Data Protection Requirements

No.	Functional Requirement
NFR-PRIV-01	The system shall ensure that personal data is collected and processed only for defined business purposes.
NFR-PRIV-02	The system shall restrict access to personal data to authorized users only.
NFR-PRIV-03	The system shall support data retention and deletion policies in accordance with regulatory requirements.
NFR-PRIV-04	The system shall protect personal data from unauthorized disclosure or misuse.

6.5. Scalability Requirements

This subsection defines the scalability requirements of the system to ensure that it can accommodate growth in users, data volume, and transaction load without impacting performance.

Table 23 Scalability Requirements

No.	Functional Requirement
NFR-SCL-01	The system shall support scaling to accommodate increasing numbers of users and transactions.
NFR-SCL-02	The system shall maintain performance levels as data volume grows.
NFR-SCL-03	The system shall support expansion to multiple tenant organizations without impacting existing tenants.

6.6. Usability Requirements

This subsection defines the usability requirements of the system to ensure that users can effectively and efficiently perform tasks within the platform.

Table 24 Usability Requirements

No.	Functional Requirement
NFR-USAB-01	The system shall provide a user-friendly interface for all supported user roles.
NFR-USAB-02	The system shall provide clear feedback and guidance for user actions and system responses.
NFR-USAB-03	The system shall minimize the number of steps required to complete key tasks such as application submission and approval.

6.7. Maintainability Requirements

This subsection defines the requirements related to the maintainability of the system, ensuring that it can be efficiently updated, configured, and supported over time.

Table 25 Maintainability Requirements

No.	Functional Requirement
NFR-MNT-01	The system shall support configuration changes without requiring major system redevelopment.
NFR-MNT-02	The system shall allow administrators to manage workflows and system parameters through configurable settings.
NFR-MNT-03	The system shall support maintainability through modular system components and clear separation of concerns.

6.8. Reliability Requirements

This subsection defines the reliability requirements of the system to ensure consistent and dependable operation under expected conditions.

Table 26 Reliability Requirements

No.	Functional Requirement
NFR-REL-01	The system shall ensure consistent processing of applications without data loss.

No.	Functional Requirement
NFR-REL-02	The system shall handle system errors and exceptions gracefully.
NFR-REL-03	The system shall maintain data integrity during system operations.

6.9. Compliance Requirements

This subsection defines the compliance requirements of the system to ensure adherence to applicable regulatory and organizational standards.

Table 27 Compliance Requirements

No.	Functional Requirement
NFR-COMP-01	The system shall support auditability of all application-related activities.
NFR-COMP-02	The system shall maintain records required for regulatory reporting and compliance.
NFR-COMP-03	The system shall enforce policies related to data handling and access control in accordance with organizational standards.

7. Business Rules

This section defines the business rules that govern system behavior and decision-making processes within the **CardFlow Credit Origination Platform**. These rules establish the conditions, constraints, and criteria that influence how applications, amendment requests, and reinstatement requests are processed throughout the system.

The rules defined in this section are applied across workflows to ensure consistency, enforce policy compliance, and support accurate decision-making.

7.1. Eligibility Rules

This subsection defines the rules that determine whether a credit application, amendment request, or reinstatement request is eligible for processing within the system.

Table 28 Eligibility Rules

No.	Functional Requirement
BR-ELG-01	A credit application shall be considered eligible for submission only if all mandatory information and required supporting documents are provided.
BR-ELG-02	An amendment request shall be allowed only for applications or accounts that are in an eligible state for modification.
BR-ELG-03	A reinstatement request shall be allowed only for credit accounts that are in a suspended or inactive status.
BR-ELG-04	Requests that do not meet defined eligibility conditions shall not proceed to further processing stages.

7.2. Submission Rules

This subsection defines the rules governing the submission of applications and requests, ensuring that all required inputs are validated before processing.

Table 29 Submission Rules

No.	Functional Requirement
BR-SUB-01	Applications and requests shall be submitted only after all required data fields are completed.

No.	Functional Requirement
BR-SUB-02	Supporting documents shall be submitted in accordance with defined format and size requirements.
BR-SUB-03	The system shall prevent submission of incomplete or invalid applications and requests.
BR-SUB-04	Each submitted application or request shall be uniquely identifiable within the system.

7.3. Approval Rules

This subsection defines the rules governing the approval process, including decision-making criteria and workflow routing conditions.

Table 30 Approval Rules

No.	Functional Requirement
BR-APR-01	Applications and requests shall be routed through defined approval workflows based on configured rules.
BR-APR-02	Approval decisions shall be made by authorized users based on assigned roles and approval authority.
BR-APR-03	Approval workflows may include multiple levels of review and approval based on defined thresholds or criteria.
BR-APR-04	Applications or requests may be approved, rejected, or returned for additional information.
BR-APR-05	Rejected applications or requests shall not proceed to further processing stages.

7.4. Amendment Rules

This subsection defines the rules specific to amendment requests and how modifications to existing records are handled within the system.

Table 31 Amendment Rules

No.	Functional Requirement
BR-AMD-01	Amendment requests shall be initiated only by authorized users.
BR-AMD-02	Amendment requests shall include updated information and supporting documentation where required.
BR-AMD-03	Approved amendments shall update the corresponding application or account records.
BR-AMD-04	Returned amendment requests may be modified and resubmitted by authorized users.

7.5. Reinstatement Rules

This subsection defines the rules governing the reinstatement of suspended or inactive credit accounts.

Table 32 Reinstatement Rules

No.	Functional Requirement
BR-RST-01	Reinstatement requests shall be initiated only for accounts that are in a suspended or inactive status.
BR-RST-02	Reinstatement requests shall include required supporting documentation.
BR-RST-03	Only authorized users shall be allowed to approve or reject reinstatement requests.
BR-RST-04	Approved reinstatement requests shall reactivate the corresponding credit account.

8. Error Handling and Exception Management

This section defines how the **CardFlow Credit Origination Platform** handles errors and exceptional conditions that may occur during system operation. It describes how validation errors, system failures, and workflow exceptions are identified, communicated to users, and managed to ensure system stability and usability.

Proper error handling ensures that users receive clear feedback, system integrity is maintained, and processing can continue or recover appropriately when issues arise.

8.1. Validation Errors

This subsection defines how the system handles errors related to invalid or incomplete user inputs during data entry and submission processes.

Table 33 Validation Errors

No.	Functional Requirement
ERR-VAL-01	The system shall detect and prevent submission of incomplete or invalid data.
ERR-VAL-02	The system shall display clear and descriptive validation error messages to users.
ERR-VAL-03	The system shall indicate the specific fields or inputs that require correction.
ERR-VAL-04	The system shall allow users to correct validation errors before resubmission.

8.2. System Errors

This subsection defines how the system handles internal errors and unexpected failures that may occur during processing or system operations.

Table 34 System Errors

No.	Functional Requirement
ERR-SYS-01	The system shall detect and handle unexpected system errors during processing.

No.	Functional Requirement
ERR-SYS-02	The system shall provide user-friendly error messages without exposing sensitive system details.
ERR-SYS-03	The system shall log system errors for monitoring and investigation purposes.
ERR-SYS-04	The system shall ensure that partial or failed operations do not result in data inconsistency.

8.3. Workflow Exceptions

This subsection defines how the system manages exceptions that occur within workflow processes, including routing failures, approval delays, or invalid workflow states.

Table 35 Workflow Exceptions

No.	Functional Requirement
ERR-WF-01	The system shall handle exceptions in workflow routing and processing.
ERR-WF-02	The system shall allow authorized users to reprocess or reroute requests when workflow exceptions occur.
ERR-WF-03	The system shall notify relevant users when workflow interruptions or issues occur.
ERR-WF-04	The system shall ensure that workflow exceptions do not result in loss of application data.

8.4. Recovery and Retry Considerations

This subsection defines how the system supports recovery from errors and retries mechanisms to ensure continuity of operations.

Table 36 Recovery and Retry Considerations

No.	Functional Requirement
ERR-RCV-01	The system shall allow retry of failed operations where applicable.
ERR-RCV-02	The system shall support recovery mechanisms to restore normal system operations after failure.
ERR-RCV-03	The system shall ensure that retried operations do not result in duplicate or inconsistent data.

9. Traceability Considerations

This section defines the approach used to maintain traceability between business requirements, system requirements, and downstream artifacts such as test cases and implementation components. Traceability ensures that all requirements are uniquely identified, consistently tracked, and verifiable throughout the system development lifecycle.

Maintaining traceability supports impact analysis, validation, and completeness of requirements across all stages of system development.

9.1. Requirement Mapping to BRD

This subsection describes how requirements are uniquely identified within the document to support traceability and reference.

Table 37 Requirement Mapping to BRD

No.	Functional Requirement
TRC-01	Each requirement defined in this document shall be assigned a unique identifier.
TRC-02	Requirement identifiers shall follow a structured naming convention based on functional areas (for example, FR-APP, FR-REV, NFR-SEC).
TRC-03	Requirement identifiers shall remain consistent throughout the system development lifecycle.

9.2. Requirement Identification Approach

This subsection defines how requirements in this document are mapped to corresponding business requirements and validation artifacts.

Table 38 Requirement Identification Approach

No.	Functional Requirement
TRC-04	Functional and non-functional requirements shall be traceable to corresponding business requirements defined in the Business Requirements Document.

No.	Functional Requirement
TRC-05	Requirements shall be mapped to test cases to ensure validation of system behavior.
TRC-06	Traceability relationships shall be maintained through a Requirements Traceability Matrix (RTM) .

9.3. Traceability Maintenance

This subsection defines how traceability is maintained throughout the lifecycle of the system.

Table 39 Traceability Maintenance

No.	Functional Requirement
TRC-07	Traceability information shall be updated when requirements are modified.
TRC-08	Traceability shall support impact analysis for changes to requirements.
TRC-09	Traceability records shall be accessible to relevant stakeholders for review and validation.